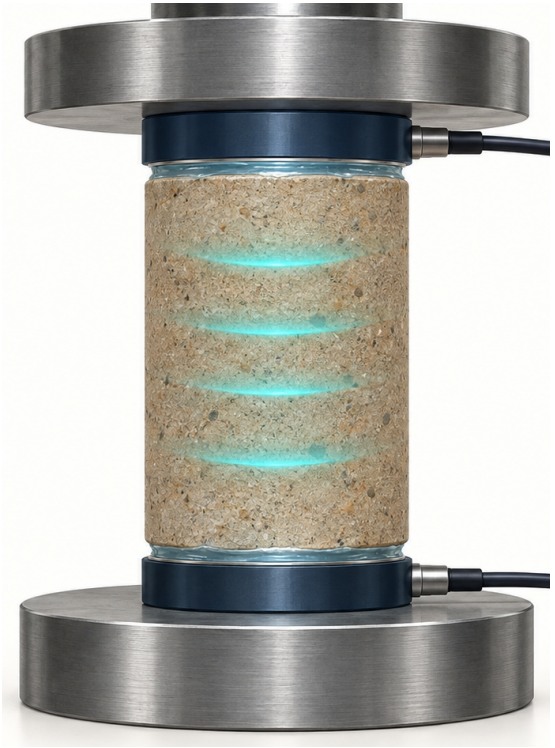
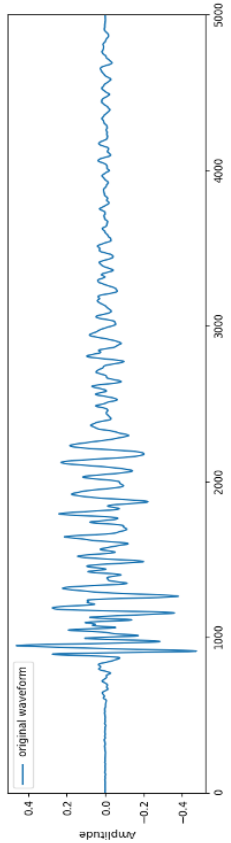
Elastic waves as rock-state probes

From velocity to dynamic moduli, cracks, fluids and damage

Petrophysics · Lecture 5





What changes inside a rock when its waveform changes?

Learning objectives and lecture sequence

By the end of this lecture, students should be able to:

01 derive dynamic elastic moduli from V_p, V_s and density

02 predict how crack closure, crack growth and saturation affect wave observations

03 separate travel-time evidence from amplitude and waveform evidence

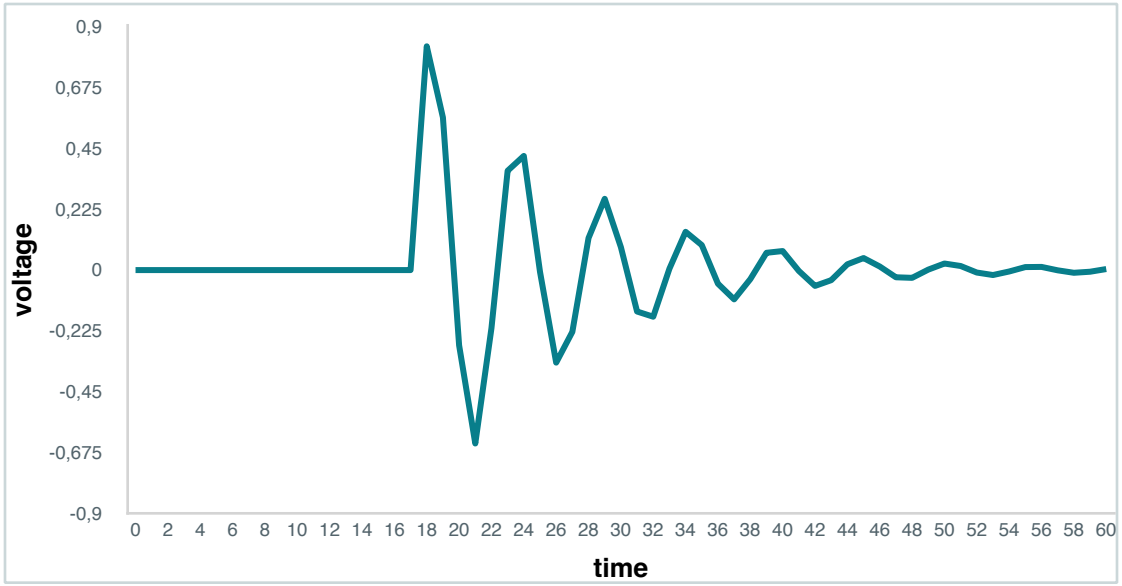
04 construct a defensible rock-state interpretation with explicit assumptions

SEQUENCE

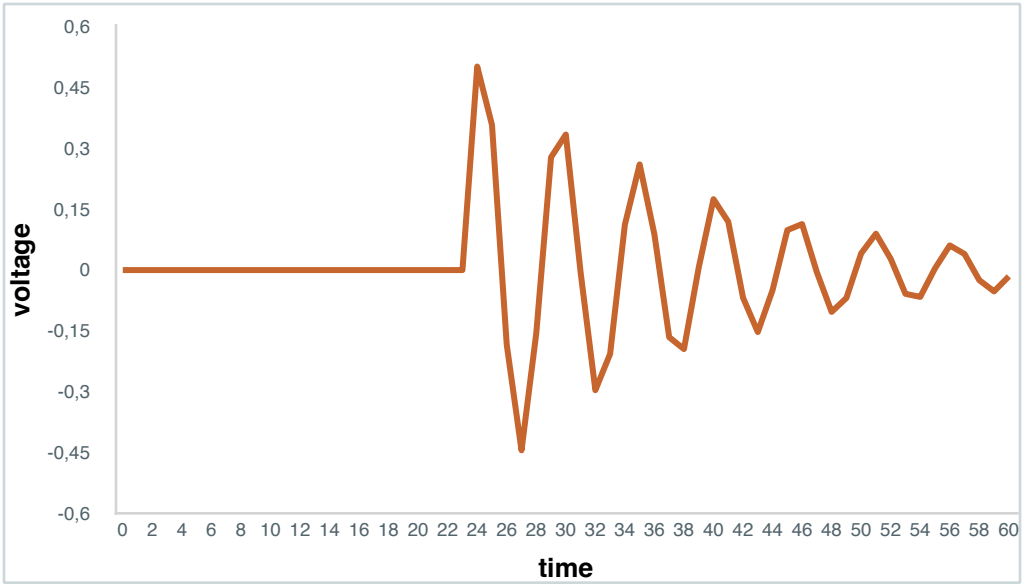
dynamic moduli → cracks and stress → fluids → attenuation → integrated diagnosis

Opening problem: two waveforms from the same specimen

The path length and electronics are unchanged. The second pulse arrives later and has a weaker, longer coda.



STATE A



STATE B

Predict one change in the rock that could produce all three observations.

Think for 30 seconds, then compare the evidence used in your explanation.

A waveform supports several levels of inference

OBSERVE arrival time · amplitude · frequency content · coda

→

CALCULATE velocity · velocity ratio · attenuation proxy · dynamic moduli

→

INFER **stiffness · cracks · saturation · anisotropy · damage**

Every downward step adds assumptions.

Wave speed depends on elastic resistance and density

P WAVE

$$V_p = \sqrt{\frac{K + \frac{4}{3}G}{\rho}}$$

sensitive to resistance against volume change and shear

S WAVE

$$V_s = \sqrt{\frac{G}{\rho}}$$

sensitive to shear resistance; absent in a pure fluid

Forward problem $K, G, \rho \rightarrow V_p, V_s$

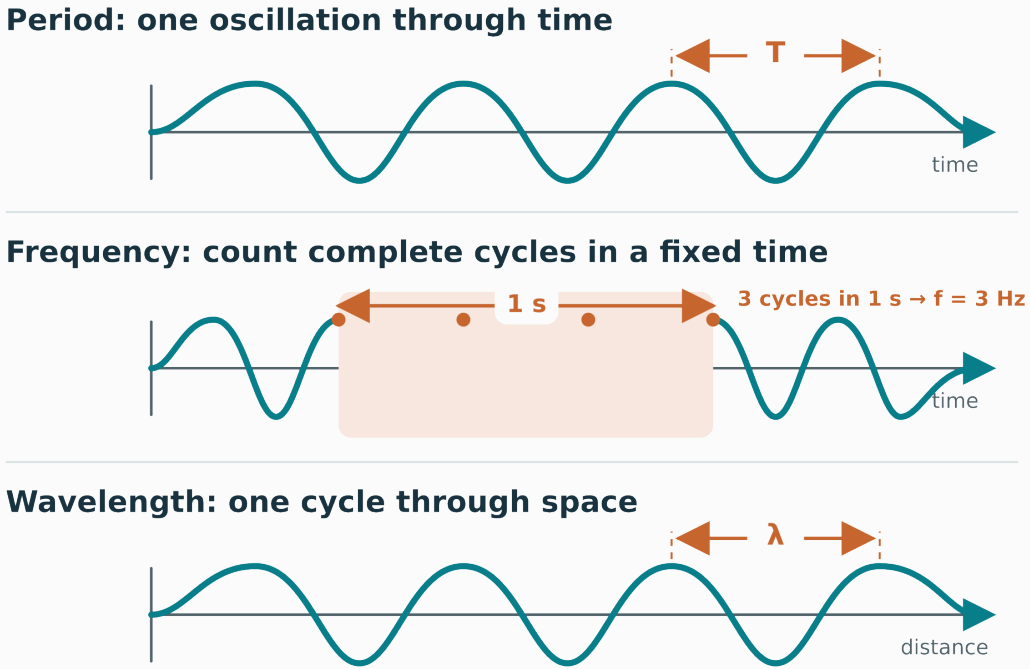
Inverse problem $V_p, V_s, \rho \rightarrow$ moduli

Frequency and wavelength describe different aspects of the wave

Quantity	Meaning	Unit
Period T	time for one oscillation	s
Frequency f	oscillations per second	Hz
Wavelength λ	distance occupied by one cycle	m
Velocity V	speed of propagation	m s^{-1}

$$f = \frac{1}{T}$$

$$V = f\lambda$$



Changing frequency changes wavelength even when the rock velocity is unchanged.

Worked example: a 1 MHz pulse in sandstone

Assume a P-wave velocity of 4 000 m s^{-1} and a dominant frequency of 1 MHz.

$$\lambda = V / f$$

$$\lambda = 4\,000 / 1\,000\,000 \text{ m}$$

$$\lambda = 0.004 \text{ m} = 4 \text{ mm}$$

Compare 4 mm with:

- grain diameter
- pore size
- crack length
- specimen diameter



Would this wave resolve individual grains, or mainly their collective effect?

A wavelength sets the scale over which the rock is averaged

HETEROGENEITY ≪ λ

Many grains or pores contribute to one wavelength. The wave responds mainly to an effective stiffness and density.

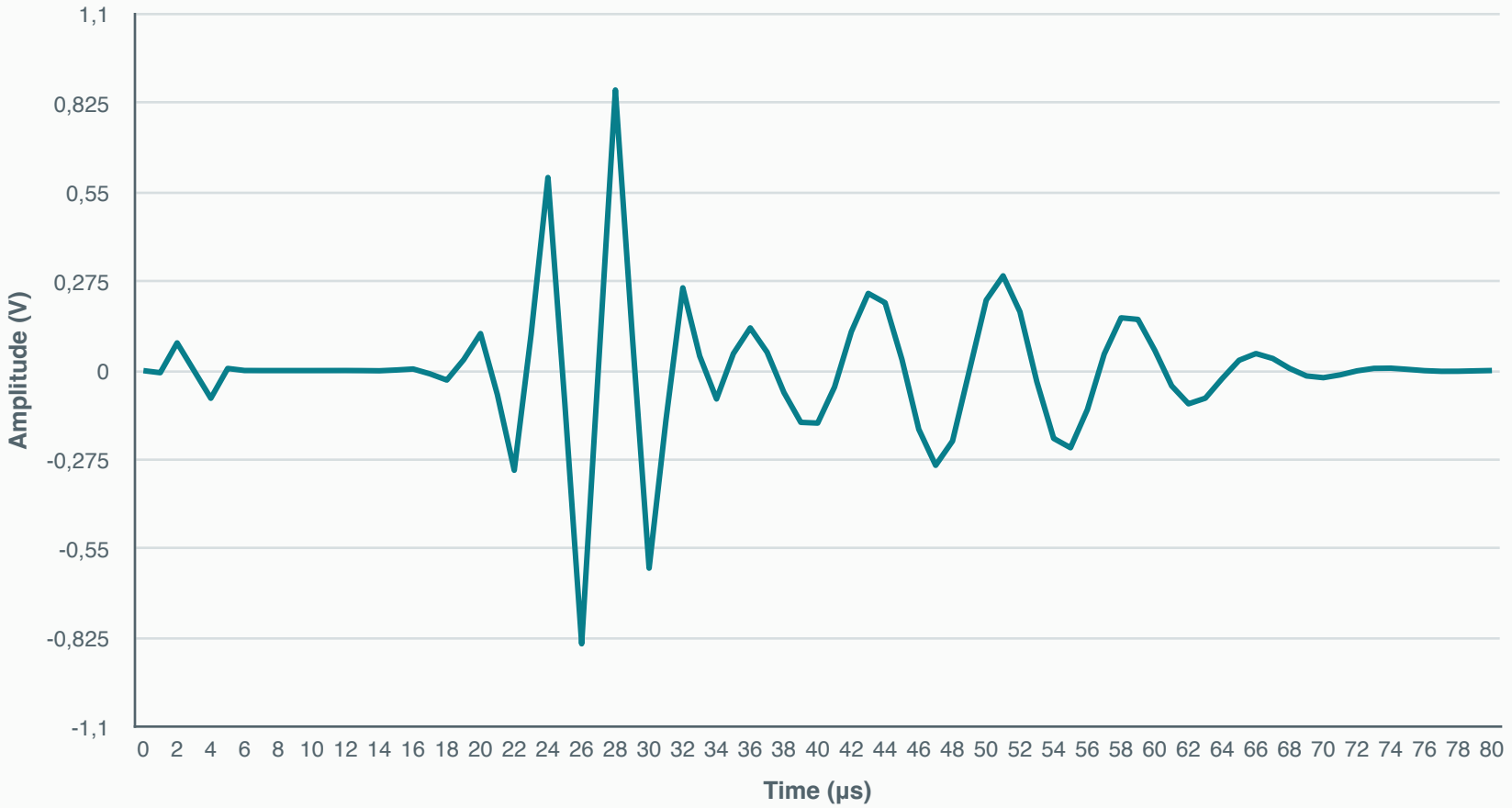
HETEROGENEITY ≈ λ

Scattering becomes important. Energy leaves the coherent first arrival and the waveform becomes more complex.

What the wave ‘sees’ depends on frequency as well as rock structure.

A different transducer frequency can produce a different measurement of the same specimen.

A received waveform contains more than one arrival



① trigger / cross-talk

② first coherent arrival

③ later cycles and coda

④ background noise

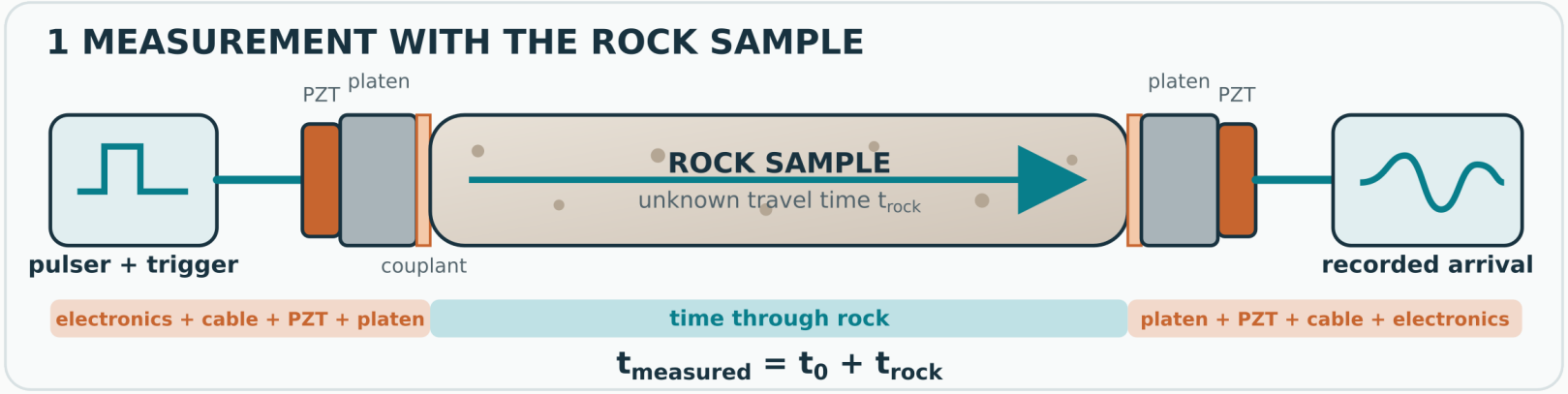
Which point would you choose as the arrival, and how would you reproduce that choice?

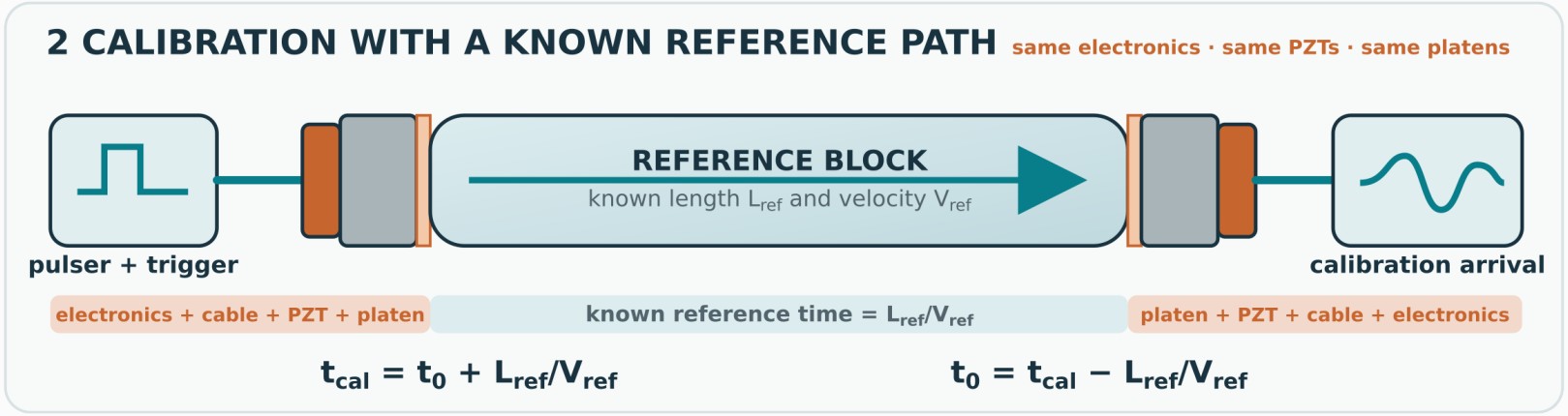
Measured time includes delays outside the rock

trigger + electronics + transducers + platens + propagation through rock = measured arrival time

t_0 must be measured using an independent calibration or a known reference path.

Ignoring t_0 makes the rock appear slower than it is.





Velocity requires corrected time and the current path length

$$v = \frac{L_{inst}}{t_{rock}}$$

L = Instantaneous propagation distance through the specimen

t_rock travel time after removing system delay

Use L_0 for strain. Use the current L for the propagation distance.

During compression

$$L_{inst} = L_0(1 - \epsilon_a)$$

Compression and shortening are positive in our sign convention. The current length decreases as ε_a increases.

Worked example: velocity through the Sandstone specimen

Use the initial specimen length L_0 = 50 mm. At the selected load, ε_a = 0.002. The measured arrival is 26 µs and the calibrated system delay is 4 µs.

Step	Calculation	Result
Current length	$L = 50(1 - 0.002)$	75.848 mm
Rock travel time	$t_{rock} = 26 - 4$	22 µs
Unit conversion	$L = 0.075848$ m; $t = 22 \times 10^{-6}$ s	SI units
P-wave velocity	$V_p = 0.075848 / (22 \times 10^{-6})$	3 448 m s^{-1}

The numbers are illustrative; the calculation structure will be applied to the experimental data.

Wave speed depends on elastic resistance and density

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Forward problem $K, G, \rho \rightarrow V_p, V_s$

Inverse problem $V_p, V_s, \rho \rightarrow$ **moduli**

Dynamic moduli from V_p, V_s and density

Shear modulus

$$G_{\text{dyn}} = \rho V_S^2$$

Bulk modulus

$$K_{\text{dyn}} = \rho \left(V_P^2 - \frac{4}{3} V_S^2 \right)$$

Young's modulus

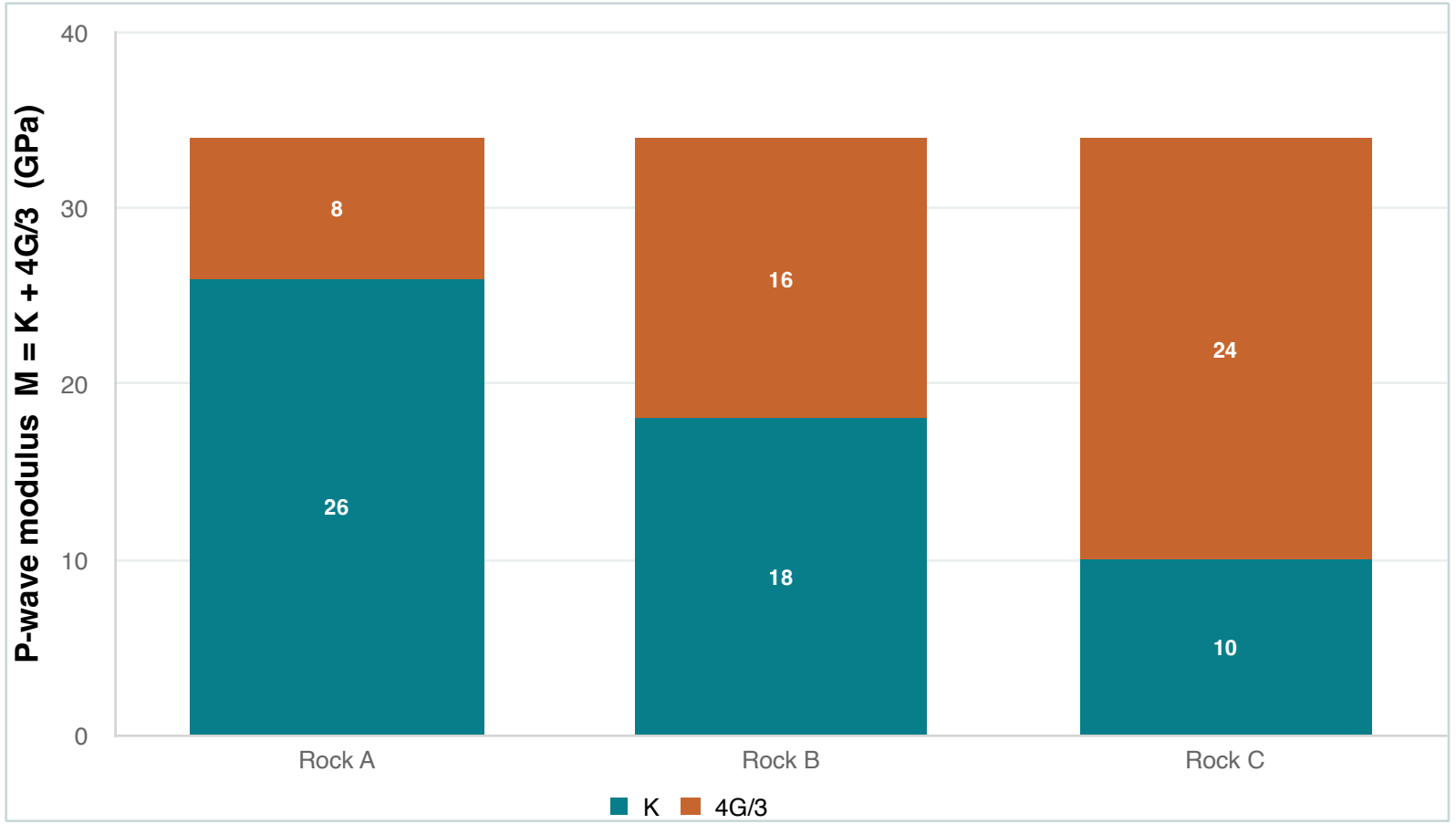
$$E_{\text{dyn}} = \rho V_S^2 \left(\frac{3V_P^2 - 4V_S^2}{V_P^2 - V_S^2} \right)$$

Poisson's ratio

$$\nu_{\text{dyn}} = \frac{V_P^2 - 2V_S^2}{2\left(V_P^2 - V_S^2\right)}$$

Use SI units before substitution: ρ in kg m^{-3} and velocity in m s^{-1}.

V_p alone cannot separate bulk and shear stiffness



All three have the same

$$M = K + \frac{4}{3}G$$

and therefore the same V_p when density is equal.

V_s supplies the independent constraint on G.

Individual calculation check

A dry rock has:

$$\rho = 2\,400 \text{ kg m}^{-3} \quad V_p = 5\,000 \text{ m s}^{-1} \quad V_s = 3\,000 \text{ m s}^{-1}$$

1. Calculate Gdyn.
2. Calculate Kdyn.
3. Estimate Edyn and Poisson's ratio.
4. State the continuum assumptions used in the inversion.

4 minutes
Show units

Calculation check: solution

GDYN	KDYN	EDYN	POISSON RATIO
21.6 GPa	31.2 GPa	52.65 GPa	0.219

Required interpretation assumptions

homogeneous at the wavelength scale isotropic linear elastic density representative of the wave path

A precise number does not remove model assumptions.

Static and dynamic moduli probe different responses

STATIC TEST

Apply a macroscopic load and measure the resulting deformation

- Modulus from the slope of the stress–strain curve
- Finite strain over a selected loading interval
- Captures the bulk response: crack closure, grain-contact deformation, frictional sliding and possible irreversible strain

DYNAMIC TEST

Send a small elastic wave through the rock and measure its velocity

- Moduli calculated from \(V_p\), \(V_s\) and density
- Very small strain and short loading time
- Probes the instantaneous elastic stiffness along the wave path, especially the stiffness of grain contacts and cracks

Why can E_{dyn} exceed E_{static}?

Static and dynamic moduli probe different responses

STATIC TEST

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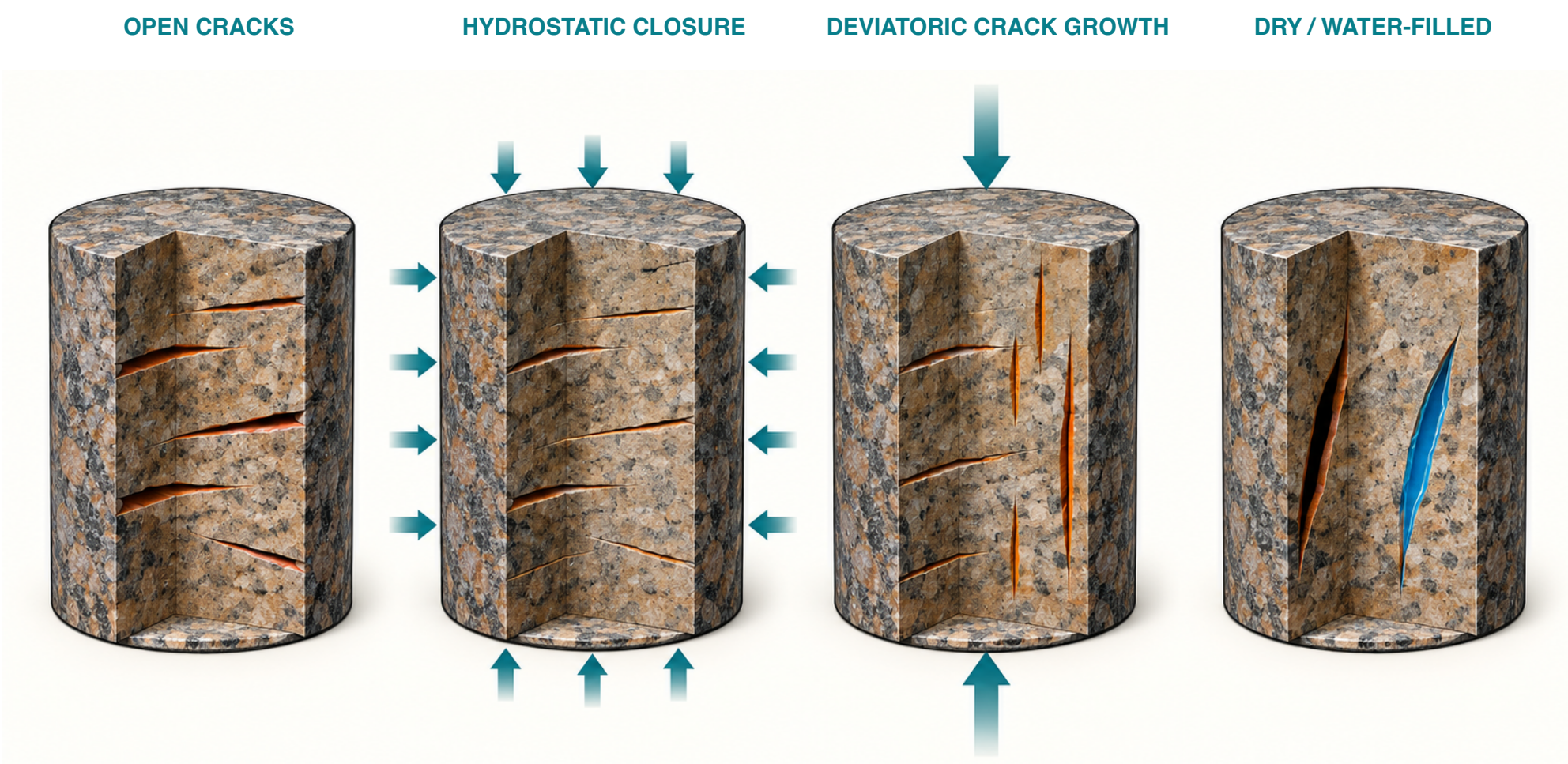
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Why can E_{dyn} exceed E_{static}?

Compliant cracks and contacts may contribute more to quasi-static deformation than to the small-amplitude ultrasonic response.

Compare values only when stress state, frequency, strain amplitude and saturation are known.

Cracks make elastic properties sensitive to rock state of stress



Thin cracks contribute little volume but large compliance.

Integrated interpretation of elastic-wave observations

Observation	Plausible rock change	Check before concluding
V_p and V_s increase	closure of compliant cracks	pressure path and saturation
transverse V_p decreases	axial crack growth	directional measurements
V_p/V_s increases	fluid-filled cracks	mineralogy and stress
amplitude falls	attenuation or scattering	coupling and gain
coda lengthens	more scattering interfaces	frequency content

These are hypotheses to test, not one-to-one translations.

From waveform to rock-state hypothesis

MEASURE		CALCULATE		COMPARE		INFER
travel time	→	V_p, V_s	→	direction	→	crack closure
amplitude		V_p/V_s		stress path		crack growth
spectrum (frequency content)		K, G, E, Poisson ratio		saturation		fluid effects

Three takeaways

- V_p and V_s together recover dynamic moduli.
- Directional changes expose crack anisotropy.
- Amplitude strengthens an interpretation only when the measurement chain is controlled.